



Technical Exam



Aptitude Exam

Vcube Aptitude Exam

Time & Work · Pipes & Cisterns · 15 reviewed questions

APTITUDE EXAM RESULT

9/15

Pass

✓ 9 Correct ✗ 6 Wrong – 0 Skipped

✓ Result sent to the separate "Aptitude" sheet.

Correct answers and reviewed explanations are shown below.

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1 TIME & WORK

Five workers earn Rs.100 each per day and finish a job in 10 days. If each earns Rs.120 per day and productivity rises by 25%, how much money is saved?

☒ A. Rs.200☐ B. Rs.250☐ C. Rs.300☐ D. Rs.500

Explanation: Original cost is $5 \times 10 \times 100 = \text{Rs.}5,000$. At 25% higher productivity the job takes $10/1.25 = 8$ days, costing $5 \times 8 \times 120 = \text{Rs.}4,800$. Saving = Rs.200.

2 TIME & WORK

A, B, and C complete work worth Rs.1,800. They work 6, 4, and 9 days, and their daily wages are in the ratio 5:6:4. How much does A receive?

☒ A. Rs.600

☐ B. Rs.750

☐ C. Rs.800

☐ D. Rs.900

Explanation: Earning weights are $6 \times 5 : 4 \times 6 : 9 \times 4 = 30:24:36 = 5:4:6$. A receives $\frac{5}{15} \times 1,800 = \text{Rs.600}$.

3 TIME & WORK

A, B, and C can complete a job alone in 11, 20, and 55 days. A works every day and is assisted by B and C on alternate days. When is the job completed?

☐ A. 7 days

☒ B. 8 days

☐ C. 9 days

☐ D. 10 days

Explanation: In two days they complete $(A+B)+(A+C) = \frac{2}{11} + \frac{1}{20} + \frac{1}{55} = \frac{1}{4}$. Four such cycles complete the work in 8 days.

4 TIME & WORK

A, B, and C can complete a job in 20, 30, and 60 days. A works daily and B and C assist together every third day. How long does the work take?

☐ A. 12 days

☒ B. 15 days

☐ C. 16 days

☐ D. 18 days

Explanation: In each three-day cycle, work done is $\frac{3}{20} + \frac{1}{30} + \frac{1}{60} = \frac{1}{5}$. Five cycles take 15 days.

5 TIME & WORK

A, B, and C can finish a job alone in 90, 40, and 12 days and work one day each in rotation, starting with A. They share Rs.240 in proportion to work performed. What are their shares?

☐ A. A Rs.24, B Rs.54, C Rs.162

☒ B. A Rs.22, B Rs.50, C Rs.168

☐ C. A Rs.26, B Rs.52, C Rs.162

☐ D. A Rs.20, B Rs.44, C Rs.176

Explanation: Using 360 work units, daily rates are 4, 9, and 30. Eight cycles complete 344 units; A and B add 13, and C completes the final 3. Contributions are 36:81:243, giving Rs.24, Rs.54, and Rs.162.

6 TIME & WORK

Food is sufficient for 400 men for 50 days. After 20 days, 100 men leave. For how many more days will the remaining food last?

☐ A. 42 days

☐ B. 50 days

☒ C. 40 days

☐ D. 60 days

Explanation: After 20 days, food remaining is $400 \times 30 = 12,000$ man-days. For 300 men it lasts $12,000/300 = 40$ days.

7 TIME & WORK

Food is sufficient for 1,500 men for 40 days at 800 g per person per day. After 20 days, 300 leave and each remaining person consumes 1,000 g daily. How many more days will the food last?

☐ A. 10 days

☒ B. 20 days

☐ C. 40 days

☐ D. 30 days

Explanation: Remaining food is $1,500 \times 20 \times 800 = 24,000,000$ g. New daily consumption is $1,200 \times 1,000 = 1,200,000$ g, so it lasts 20 days.

8 TIME & WORK

A contractor plans 124 days with 120 workers. After 64 days, two-thirds of the work is complete. How many workers can be discharged and still finish on time?

☐ A. 40

☐ B. 50

☐ C. 48

☒ D. 56 workers

Explanation: At the observed rate, the remaining one-third needs half of the 64-day effort: 120×32 worker-days. Spread over 60 days, 64 workers are required, so $120 - 64 = 56$ may leave.

9 TIME & WORK

A job was planned for 56 days with 104 people working 8 hours daily. After 30 days only two-fifths is complete. With 26 days left and 9-hour days, how many additional people are needed?

☐ A. 160☐ B. 150☒ C. 24☐ D. 56 additional people

Explanation: The remaining work is $\frac{3}{2}$ times the completed work: $(\frac{3}{2}) \times 104 \times 8 \times 30$ person-hours. Dividing by 26×9 gives 160 total people, so 56 additional people are required.

10 TIME & WORK

A contractor employs 60 workers for a 90-day project. After 60 days, three-fourths is complete. How many workers can be discharged and still finish exactly on time?

☐ A. 40☐ B. 20☒ C. 30☐ D. 15

Explanation: The remaining one-fourth equals one-third of the completed three-fourths, requiring $60 \times 60 / 3 = 1,200$ worker-days. Over 30 days this needs 40 workers, so 20 can be discharged.

11 PIPES & CISTERNS

Pipe A fills a tank in 4 hours. Together, A and B fill it in 2.5 hours. How long does B alone take?

☒ A. $6 \frac{3}{8}$ hours☐ B. $6 \frac{2}{3}$ hours☐ C. $5 \frac{1}{6}$ hours☐ D. 5 hours

Explanation: B's rate is $1/2.5 - 1/4 = 2/5 - 1/4 = 3/20$ tank per hour. Therefore B takes $20/3 = 6\frac{2}{3}$ hours.

12 PIPES & CISTERNS

Pipes A and B together fill a cistern in 4 hours. B alone takes 6 hours longer than A. How long does A alone take?

☐ A. 1 hour

☒ B. 2 hours

☐ C. 6 hours

☐ D. 8 hours

Explanation: Let A take x hours and B $x+6$. Then $1/x + 1/(x+6) = 1/4$, giving $x^2 - 2x - 24 = 0$ and the positive root $x=6$.

13 PIPES & CISTERNS

Pipes A and B together fill a tank in $6\frac{2}{3}$ minutes. A takes 3 minutes longer than B. How long does A take to fill one-third of the tank?

☐ A. 4.5 minutes

☐ B. 6 minutes

☒ C. 5.5 minutes

☐ D. 5 minutes

Explanation: The combined rate is $3/20$. Solving $1/x + 1/(x+3) = 3/20$ gives $B=12$ and $A=15$ minutes. A fills one-third in $15/3 = 5$ minutes.

14 PIPES & CISTERNS

Pipes A and B fill a tank in 24 and 32 minutes. Both start together, but B is closed early so the tank fills in 18 minutes. After how many minutes should B be closed?

- ☒ A. 8 minutes
- ☐ B. 6 minutes
- ☐ C. 12 minutes
- ☐ D. 10 minutes

Explanation: A runs for 18 minutes and fills $18/24 = 3/4$. B must fill the remaining $1/4$, requiring $32/4 = 8$ minutes.

15 PIPES & CISTERNS

Pipes A and B fill a tank alone in 36 and 48 hours. If both remain open together, how long will they take to fill the tank?

- ☒ A. $20 \frac{4}{7}$ hours
- ☐ B. $20 \frac{1}{2}$ hours
- ☐ C. 24 hours
- ☐ D. 27 hours

Explanation: Their combined rate is $1/36 + 1/48 = 7/144$ tank per hour, so time is $144/7 = 20 \frac{4}{7}$ hours. The PDF's highlighted 27-hour answer is mathematically incorrect and its original stem was incomplete.